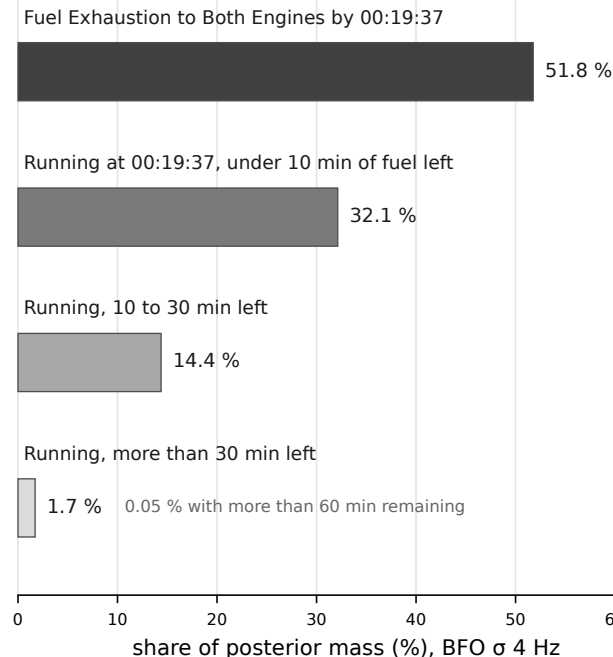
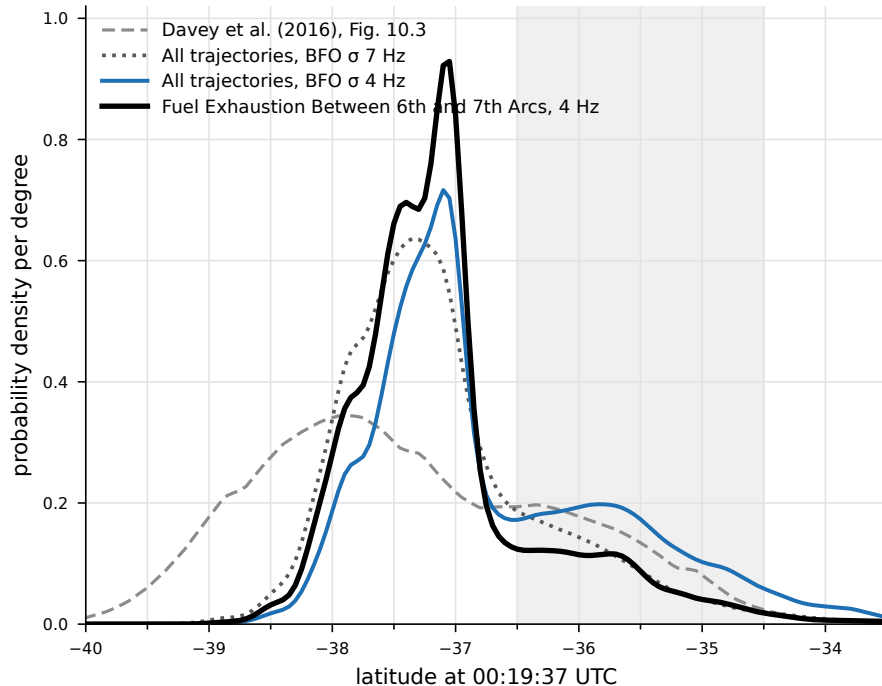




Fuel exhaustion (FE) is an OUTPUT of this model, not an input. The 00:17:30 exhaustion-time term is absent, so the share of trajectories reaching FE between the 6th and 7th arcs is a result rather than a condition imposed on the ensemble. Selecting that subset is therefore a post hoc selection.

The only difference between the two models is the BFO measurement standard deviation, 7 Hz against 4 Hz; the 4 Hz value is close to the 4.32 Hz of Davey et al. (2016). Tightening it improves log evidence from -98.420 to -96.901 , a gain of 1.52 nats, which is far outside the 0.34-nat spread of the 7 Hz run's own eight per-seed values. It moves the median 0.22° further north and raises the shoulder mass from 0.174 to 0.293, from 72 % to 122 % of the 0.241 of Davey et al. So the northern shoulder is substantially a BFO-width effect.

CONVERGENCE. The 4 Hz run has four replicates, so its split-half overlap is quoted against the four-replicate floor of 0.896, not the 0.924 that applies at eight: it averages 0.857 over the three balanced partitions, range 0.818 to 0.891, and FAILS. At matched replicate count the two runs are comparable — the first four seeds of the 7 Hz run give 0.843 — so 4 Hz is no harder to sample. The shoulder mass and the mode probabilities are the quantities that carry the failure and must not be quoted as converged. Measured on this run's own three partitions, the median agrees between halves to 0.029° on average and 0.038° at worst, so it is quotable; the 50 % interval width agrees only to 0.25° and the shoulder mass to 0.021, so neither is.

FE before 00:11:00 is excluded by the data rather than assumed away: the aircraft transmitted at 00:11, so any trajectory already dry carries a log-likelihood penalty of -50 , a weight of about 2×10^{-22} . FE time is stored at 128 s resolution, so only about 21 times are representable; the single slot spanning 00:09:36–00:11:44 straddles the 6th arc, and mass in it is a storage artefact rather than a distinct category. FE time over all trajectories — recorded where FE occurs by 00:19:37, projected from each trajectory's own mean burn rate where it does not — has median 00:19:12, mean 00:21:38 and a 95 % interval of 00:10:40 to 00:46:18 at 4 Hz, against 00:19:12, 00:20:13 and 00:10:40 to 00:41:36 at 7 Hz.

Left panel: median latitude -37.00° over all trajectories at 4 Hz and -37.19° over those reaching FE between the arcs, with 50 % intervals of 0.95° and 0.65° . Right panel, 4 Hz: 98.3 % of the posterior is within 30 minutes of FE at the time of the final transmission, and 83.9 % within 10 minutes.